

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	AT1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
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Date:	21-08-2026	Duration:	60 minutes

CO5 Statement: Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly.

Code of Conduct

I Y Praveen Raj (192372306) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sunny (Praveen)

Assigned Problem Statement

Formal Verification Framework for E-Transaction Protocols Using Product Automata and Reachability Analysis for Fraud Detection

1. Introduction and Scope of the Survey

Electronic transaction (e-transaction) systems — covering card-present and card-not-present payments, contactless (NFC) transfers, and blockchain-based settlement — are exposed to a recurring set of attack classes: replay attacks, double-spending, identity forgery, and race conditions arising from concurrent transaction processing. This literature survey supports the assigned problem statement by examining how existing research verifies the correctness and fraud-resistance of e-transaction protocols, with particular attention to automata-based formal modelling and reachability analysis as the mechanism for exhaustively exploring protocol states. Five research works were selected to represent the two dominant methodological families in this space: (i) formal-methods approaches that build automata or symbolic models of a protocol and use model checking or reachability analysis to prove or disprove security properties, and (ii) data-driven approaches that use statistical or graph-based machine learning over transaction logs to flag fraudulent behaviour after the fact. Studying both families together makes the research gap — and the motivation for a product-automata, reachability-based framework — explicit.

2. Individual Paper Reviews**2.1 Fedotov & Khritankov (2021) — Statistical Model Checking of Common Attack Scenarios on Blockchain**

Objective: To verify blockchain transaction-processing systems against three concrete attack scenarios — DNS spoofing, double-spending via memory-pool (mempool) flooding, and consensus-delay attacks — using formal verification rather than ad-hoc testing.

Methodology & Technologies: The authors build a formal model for each attack scenario and apply statistical model checking, which repeatedly samples model executions to estimate the probability that a temporal-logic property is violated, following a modelling → running → analysis loop with counterexample-guided refinement.

Dataset/Tools: A statistical model checker driven by probability distributions calibrated from historical Bitcoin network telemetry (mempool size, propagation and confirmation delay).

Key Findings: The approach quantifies attack-success probability as a function of network parameters and proposes concrete mitigations (e.g., mempool size limits, confirmation-count thresholds) for each attack class.

2.2 Kang, Woo & Hong (2023) — Bitcoin Double-Spending Attack Detection using Graph Neural Network

Objective: To detect double-spending attempts in a Bitcoin-like network using only partial visibility from a small set of observer nodes.

Methodology & Technologies: Network topology and mempool membership are represented as a graph; a graph neural network (GNN) is trained to predict whether a given transaction is present across all nodes' mempools, inferring double-spending from inconsistent propagation patterns.

Dataset/Tools: A simulated peer-to-peer Bitcoin network topology in which roughly 1% of nodes act as labelled observers; a deep-learning framework for GNN training.

Key Findings: The GNN classifier achieves at least 0.95 detection accuracy even with minimal observer coverage, showing that statistical/learning methods can detect double-spending without full formal modelling of the underlying protocol.

2.3 A Survey of Practical Formal Methods for Security (Formal Aspects of Computing, 2022)

Objective: To synthesise how formal methods (FM) have been applied across security-critical domains, including electronic trading systems and contactless/EMV payment protocols.

Methodology & Technologies: The survey catalogues the use of model checkers (Scyther, FDR, NuSMV, CL-Atse) and automata-learning techniques to build models of standards such as EMV/NFC payment protocols, and timed automata combined with timed temporal logic to model application behaviour for intrusion and fraud detection.

Dataset/Tools: Case studies drawn from published verification efforts on NFC payment protocols and industrial control system logs, rather than a single unified dataset.

Key Findings: Automata-based modelling of payment protocols and lightweight formal specification for fraud detection can reach up to 100% attack-detection rates on log data in reported case studies, but require substantial manual protocol-modelling effort and are typically scoped to a single protocol variant rather than a composable, reusable automaton family.

2.4 Khaleghpour (2026) — Leakage-Safe Graph Features for Interpretable Fraud Detection in Temporal Transaction Networks

Objective: To detect illicit transactions in temporal transaction networks using interpretable, causally-computed graph features rather than opaque end-to-end models.

Methodology & Technologies: A time-respecting (causal) graph-construction protocol restricts feature computation to edges observed up to each timestep, avoiding look-ahead leakage; structural descriptors include degree statistics, PageRank, HITS hub/authority scores, k-core indices, and two-hop reachability proxies.

Dataset/Tools: The Elliptic dataset, a labelled Bitcoin transaction graph distinguishing licit and illicit entities; evaluation via ROC-AUC, Average Precision, and Precision-at-K to reflect realistic investigator triage.

Key Findings: Interpretable, leakage-safe structural features achieve competitive fraud-detection performance and, notably, use reachability-style graph proxies as a detection signal — but as a statistical heuristic rather than an exhaustive formal guarantee.

2.5 Wright (2025) — Safe Low Bandwidth SPV: A Formal Treatment of Simplified Payment Verification Protocols and Security Bounds

Objective: To provide a complete formal specification and security-bound proof for Simplified Payment Verification (SPV), the lightweight transaction-verification method used by resource-constrained clients.

Methodology & Technologies: The SPV protocol is reconstructed from first principles and modelled using symbolic automata, Merkle-membership relations, and chain-of-proof dominance predicates, combined with game-theoretic and complexity-theoretic reasoning to bound adversarial success.

Dataset/Tools: A purely formal/analytical treatment — no empirical transaction dataset; verification is carried out through proof construction over the symbolic automaton model.

Key Findings: SPV is shown to be secure under bounded adversarial assumptions and scalable for verifiable transaction inclusion, correcting earlier informal claims about its weaknesses; the paper demonstrates automata-based reasoning applied to a single, narrowly scoped verification protocol rather than to the broader set of fraud classes together.

3. Comparative Literature Survey Table

Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitations	Research Gap
Fedotov & Khritankov (2021)	Verifying blockchain systems against DNS, double-spend & consensus-delay attacks	Statistical model checking; model→run→analyse loop with counterexample refinement	Statistical model checker; historical Bitcoin mempool/delay telemetry	Quantifies attack-success probability; proposes per-attack mitigations	Single-protocol, per-attack models; not reusable across attack types	No unified, composable model spanning multiple fraud classes
Kang, Woo & Hong (2023)	Detecting Bitcoin double-spending from partial network visibility	Graph neural network over mempool-membership graph	Simulated P2P topology; ~1% observer nodes	≥95% detection accuracy with minimal observer coverage	Learned model gives no formal correctness guarantee; opaque decision logic	No exhaustive state-space guarantee; detection is probabilistic, not provable
Survey of Practical FM for Security (2022)	Applying formal methods to e-trading & NFC/EMV payment protocols	Model checking (Scyther, FDR, NuSMV, CL-Atse); automata learning; timed automata + temporal logic	Case studies from published NFC/EMV verification efforts & ICS logs	Up to 100% attack detection on log data in reported case studies	Heavy manual modelling effort; each model scoped to one protocol variant	Lacks a generalised, reusable product-automaton framework for multiple protocols
Khaleghpour (2026)	Illicit-transaction detection in temporal transaction networks	Leakage-safe causal graph feature extraction (degree, PageRank, HITS, k-core, reachability proxies)	Elliptic labelled Bitcoin transaction graph	Competitive fraud detection; reachability used as a statistical feature	Reachability is a heuristic signal, not an exhaustive formal check	No integration of formal reachability analysis with graph-based fraud signals
Wright (2025)	Formal security bound for Simplified Payment Verification (SPV)	Symbolic automata; Merkle-membership relations; game-/complexity-theoretic proof	Analytical/proof-based; no empirical dataset	Proves SPV secure under bounded adversarial assumptions	Scoped to one verification protocol; does not cover replay, forgery, or race conditions	No single automaton family covers replay + double-spend + forgery + race conditions together

4. Limitations and Challenges of Existing Approaches

- Narrow protocol scope: formal/automata-based works (2.3, 2.5) each verify one protocol variant (NFC/EMV or SPV) rather than a shared framework applicable across payment channels.
- High manual modelling cost: constructing accurate automata or symbolic models for each new protocol requires significant expert effort, limiting scalability to new fraud scenarios.
- No formal guarantee in ML-based detectors: the GNN- and graph-feature-based approaches (2.2, 2.4) offer strong empirical accuracy but cannot certify the absence of an attack class — they can only flag statistically anomalous instances.
- Single-attack-class focus: most works target one fraud category at a time (double-spending, or DNS spoofing, or SPV correctness) instead of jointly covering replay attacks, double-spending, identity forgery, and race conditions.
- Reachability used inconsistently: reachability appears either as a formal exhaustive technique (patents/timed-automata literature) or as an informal graph heuristic (2.4), with no work combining both rigorously for fraud detection.

5. Identified Research Gap

Across the reviewed literature, formal-methods research demonstrates that automata and reachability analysis can exhaustively verify protocol-level security properties, but each study is scoped to a single protocol and requires bespoke modelling. Data-driven research demonstrates that statistical and graph-based signals — including reachability-style proxies — are effective for fraud detection at scale, but these signals are heuristic and offer no completeness guarantee. No reviewed work constructs a product-automaton model that composes multiple per-protocol or per-attack automata into a single verifiable system and then applies systematic reachability analysis to certify coverage of replay attacks, double-spending, identity forgery, and race conditions simultaneously. This is precisely the gap the assigned problem statement — a Formal Verification Framework for E-Transaction Protocols Using Product Automata and Reachability Analysis for Fraud Detection — addresses.

6. Future Research Directions

- Develop a product-automaton construction method that composes independently-verified per-protocol automata (card, NFC, blockchain) into one state space without a combinatorial blow-up in reachability computation.
- Integrate CTL/LTL-based reachability verification with interpretable graph features (as in 2.4) so that formally certified states can guide, and be cross-checked against, statistical fraud scores.
- Extend statistical model checking (as in 2.1) from single-attack scenarios to multi-attack composite models covering replay, double-spend, forgery, and race conditions together.
- Benchmark formal-verification coverage against ML-based detectors (2.2, 2.4) on shared datasets such as Elliptic to quantify the practical fraud-detection gain of exhaustive reachability analysis over sampling-based methods.
- Investigate automata-learning techniques to reduce the manual modelling cost identified as a limitation in the formal-methods survey (2.3), enabling faster onboarding of new transaction protocols into the framework.

7. References

- [1] I. Fedotov and A. Khritankov, “Statistical Model Checking of Common Attack Scenarios on Blockchain,” arXiv:2109.02803, 2021.
- [2] C. Kang, J. Woo, and J. W. Hong, “Bitcoin Double-Spending Attack Detection using Graph Neural Network,” arXiv:2304.13935, 2023.

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- [4] H. Khaleghpour, “Leakage Safe Graph Features for Interpretable Fraud Detection in Temporal Transaction Networks,” arXiv preprint, 2026.
- [5] C. S. Wright, “Safe Low Bandwidth SPV: A Formal Treatment of Simplified Payment Verification Protocols and Security Bounds,” arXiv:2507.00740, 2025.